

TSN ETHERNET SWITCH QSXG12

The QBIT TSN Ethernet Switch (QSXG12) is a managed Time-Sensitive Networking (TSN) switch designed for deterministic Ethernet networks. It provides 4 x 10GBASE-SFP ports and 8 x 10/100/1000BASE-T(X) TSN ports along with a console interface, enabling high-bandwidth aggregation and endpoint connectivity in converged industrial and mission-critical networks.

The switch integrates key IEEE 802.1 TSN capabilities for precise time synchronization, scheduled traffic, per-stream policing and frame replication/elimination for reliability. Comprehensive Layer-2 and Layer-3 features, along with secure configuration and lifecycle management, simplify deployment in safety and mission-critical environments.

Introduction

Time-Sensitive Networking (TSN) extends standard Ethernet with deterministic latency, bounded jitter and precise time synchronization. The TSN Ethernet Switch forms the backbone of TSN domains, providing deterministic forwarding between TSN endpoints while maintaining standard Ethernet interoperability for best-effort traffic.

It supports IEEE 802.1AS gPTP timing, IEEE 802.1Qbv time-aware scheduling, IEEE 802.1Qci per-stream filtering and policing and IEEE 802.1CB frame replication and elimination for reliability (FRER). Integrated Layer-3 routing, multicast control, redundancy mechanisms and secure management interfaces enable deployment in converged networks with mixed TSN and non-TSN traffic.



Architecture

This TSN Ethernet switch delivers deterministic, low-latency communication with 8 x 1G ports and 4 x 10G ports. The 10G ports support both optical and electrical (DAC) connections via flexible SFP/SFP+ modules. All 12 ports connect through high-speed SerDes lanes to a central Assembler/Deassembler block. The architecture splits into two main parts: the TSN data-plane subsystem and an integrated ARM/RISC-V CPU subsystem. The TSN engine features ingress classification, PSFP policing, FRER redundancy, IEEE 1588 PTP timestamping, and advanced QoS. A unified shared memory pool and non-blocking switching fabric handle traffic across all ports efficiently. Deterministic behavior is ensured via time-aware shaping (802.1Qbv), frame preemption, and automated scheduled frame injection. On egress, frames undergo modification, filtering, and precise scheduling before transmission. The CPU subsystem includes timers, sensors, eMMC, DDR memory, Frame DMA, and standard interfaces (UART, SPI, GPIO, JTAG). Overall, the design combines high bandwidth, full TSN feature set, and flexible physical-layer options for converged real-time networks.

Why Choose Us?

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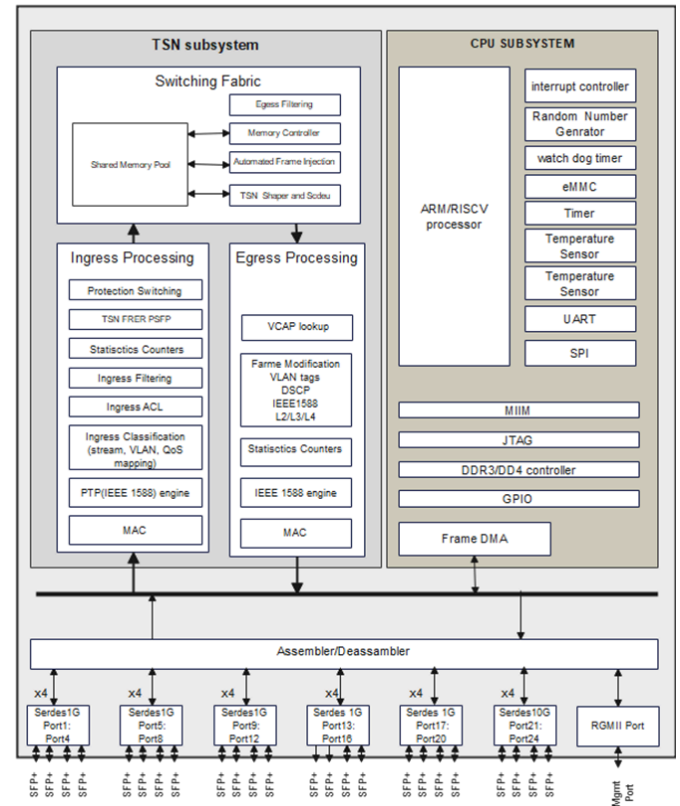
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Features

- 4 x 10GBASE-SFP TSN Ethernet ports.
- 8 x 10/100/1000BASE-T(X) TSN Ethernet ports.
- 1 x Console port; power supply input range 100-240VAC.
- Switching capacity minimum 100 Gbps with wire-speed forwarding for 64-9216 byte frames; jumbo frame support.
- IEEE 802.1AS timing and synchronization (gPTP).
- IEEE 802.1Qbv Time Aware Shaper (TAS) with scheduled traffic; status monitoring of TSN streams.
- IEEE 802.1Qci per-stream filtering and policing; IEEE 802.1CB Frame Replication and Elimination for Reliability (FRER).
- IEEE 802.3-2008 compliant Ethernet switching with automatic MAC address learning/aging and static MAC table.
- VLAN support: port-based VLANs, priority tagged/untagged frames and IEEE 802.1Q VLAN tagging (up to 4K VLAN groups) with shared or independent VLAN learning.
- QoS support: IEEE 802.1p CoS/QoS and Layer-3/4 QoS classification (e.g., DSCP) for best-effort IP traffic.
- Layer-3 features: static IPv4/IPv6 routes, OSPF and RIP, SVIs per VLAN, inter-VLAN routing and port redirection.
- Multicast support: IGMP snooping (v1/v2/v3) for IPv4, MLD snooping for IPv6, Layer-2 multicast filtering and support for at least 255 multicast groups.
- High availability: IEEE 802.3ad link aggregation (LACP), VRRP (TSN-compatible) and MTBF not less than 100,000 hours.
- Secure management: HTTPS web interface, SSHv2 CLI, SNMPv3, Syslog eventing, DHCP support, and encrypted & digitally signed firmware/bitstream upgrades with save/restore configuration.



Specifications

SN	Specification	Details
1.	Ports configuration	• 4 x 10GBASE-SFP TSN Ethernet ports • 8 x 10/100/1000BASE-T(X) TSN Ethernet ports • 1 x Console Port • Power supply voltage range 100-240VAC (or adaptor to be provided) • 1 PPS in/out • 10 MHz in/out
2.	Performance specification	• Switching capacity: minimum 100 Gbps • Wire-speed performance across all packet sizes from 64-9216 bytes • Jumbo frame support
3.	TSN features	• IEEE 802.1AS - Timing and synchronization (gPTP) • IEEE 802.1Qbv - Time Aware Shaper (TAS) • IEEE 802.1Qci - Per-stream filtering and policing • IEEE 802.1CB - Frame Replication and Elimination for Reliability (FRER) • Status monitoring of TSN stream



Specifications Cont..

4.	High availability	• Link Aggregation protocol IEEE 802.3ad (LACP) • VRRP must work together with TSN features • MTBF not less than 100,000 hours
5.	Layer 2 general functionalities	• IEEE 802.3-2008 (Ethernet) • Automatic MAC address learning and aging; static MAC table • Port-based VLANs; IEEE 802.1Q VLAN tagging (up to 4K VLAN groups) • IEEE 802.1p CoS/QoS; IEEE 802.1AB (LLDP) • Storm control for flooded broadcast, multicast and unicast • Spanning Tree Protocol: IEEE 802.1D (STP), IEEE 802.1w (RSTP), IEEE 802.1s (MSTP) • Tagged, priority tagged and untagged frame handling with configurable VLAN tag insertion • Shared and independent VLAN learning
6.	Layer 3 general functionalities	• Static IPv4 and IPv6 routes • Dynamic routing protocols: OSPF, RIP • Switched Virtual Interfaces (SVIs) for each VLAN • Inter-VLAN routing • Port redirection
7.	Multicast	• Support for at least 255 multicast groups • IGMP snooping (v1, v2 and v3) for IPv4 • MLD snooping for IPv6 • Layer 2 multicast filtering • IGMP snooping (up to 255 multicast filters)
8.	Operating Temperature	• Operating Temperature: -35°C to +45°C • Storage Temperature: -40°C to +85°C • Humidity: 10% to 90% non-condensing
9.	QoS	• Layer 3 and Layer 4 QoS classification (e.g., using DSCP) for non-TSN, best-effort IP traffic in addition to Layer 2 TSN scheduling
10.	Access Control Lists (ACLs)	• Layer 3 ACLs to filter traffic based on incoming port (logical/physical) and IP address, destination IP address, source/destination TCP/UDP port, 802.1p priority bits, ToS precedence bits, DSCP, protocol type, UDP/TCP socket, egress port, source/destination MAC address, VLAN ID and EtherType field
11.	Configuration & Management	• HTTPS web interface • SSHv2 command line interface (CLI) • Encrypted and digitally signed firmware/bitstream upgrades • Saving and restoring configuration • Internal status monitoring and logging • Event notification through Syslog and Statistics independent per port • SNMPv3 and DHCP support
12.	Dimensions	• Form Factor: 19-inch, 1U rack-mountable • Dimensions (H × W × D): 44.45 mm × 482.6 mm × 310 mm



Deliverables

1. TSN SWITCH Hardware - QSXG12
2. 4 -10G Base R SFP+ and 8 RJ45 Cables.

Get in touch with us

Delivering Advanced Solutions for
Next-Gen Networking



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